

{Visual Art Interpreter}

Deep Learning for Aesthetic Score Calibration

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Is this art piece aesthetic?

and why?

Composition

Lighting

Color

Coherence

Aesthetic quality assessment of visual art is a challenging problem at the intersection of **computer vision** and **human perception**.



Subjective

No single ground truth — two experts can rate the same painting differently



Multidimensional

One score hides independent signals: composition, lighting, color, and content



Cross-Domain

Criteria for photographs differ fundamentally from fine art paintings

Goal: Move beyond a single number – explain **WHY** an artwork scores the way it does.

{Task Definition & Proposed Method}

TWO-STAGE PIPELINE



A quantitative scoring model that predicts aesthetic quality on a 1-10 scale from image features



An explainability layer that generates natural-language descriptions of why an artwork receives its score.

Our approach combines supervised deep learning on curated aesthetic datasets with a generative vision language model (VLM) component.

The broader goal is a system that can evaluate both photographs and fine-art paintings in a style-aware manner, producing outputs accessible to non-expert users.

{Experimental Setup}

Three datasets - unified to a common 0-1 score scale:

AVA

~250k photographs
Overall score only

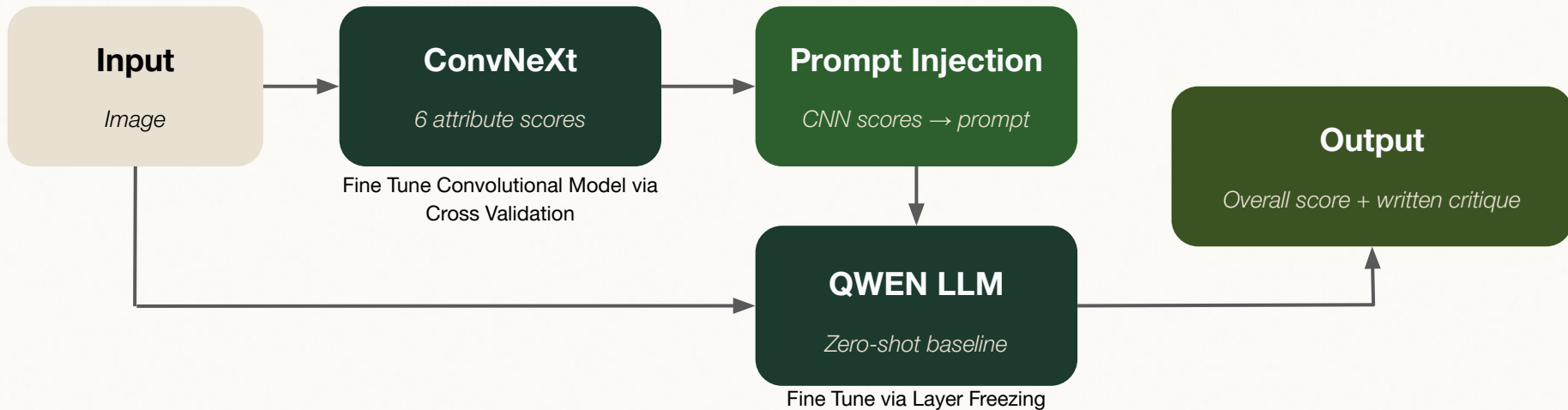
AADB

~10k photographs
Light • Color • Composition • Content

APDDv2

~1.6k paintings
Light • Color • Composition • Content

ARCHITECTURE



{Results}

LoRA Fine-Tuning vs Zero-Shot Baseline (Qwen3-VL-8b-Instruct)

Config	Description	MAE ↓	RMSE ↓	Bias ↓	Pearson ↑	±1.0	±2.0
Base	Zero-shot	1.736	2.037	1.602	0.503	29.5%	69.0%
A	Language-only LoRA	0.736	1.021	0.252	0.717	81.5%	94.8%
B	Connector-only	0.775	1.096	0.150	0.689	78.0%	95.2%
C	Vision+Connector	2.980	3.690	2.867	-0.057	23.1%	49.9%
D	Language+Vision	0.714	0.998	0.222	0.732	83.0%	95.0%



57.6%

MAE reduction
Base → Config A



84%

Bias reduction
Base → Config D



Config C

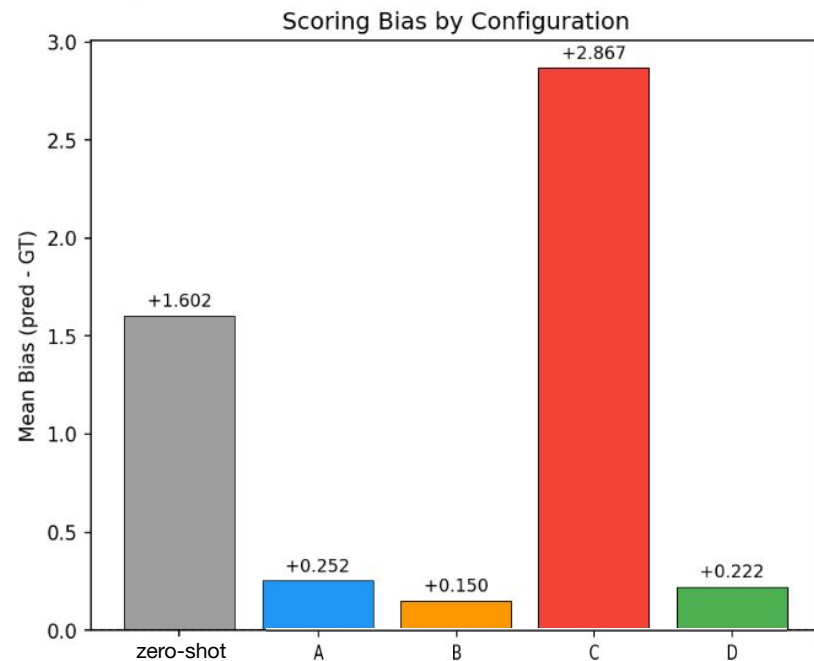
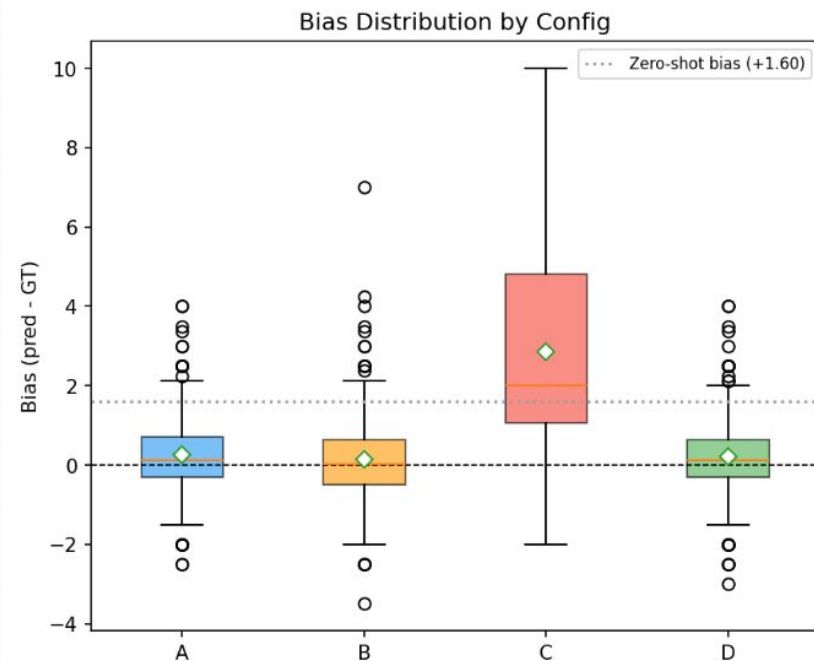
Vision+Connector
Collapsed (MAE 2.98)



0.023

MAE difference
A vs D (vision adds
negligible benefit)

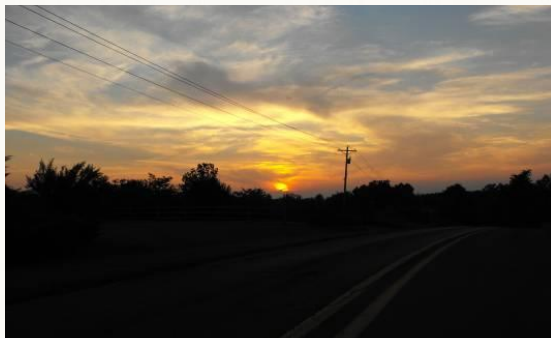
{Results}



Bias: The Closer to 0 the Better

Zero-Shot	A	Language Only	B	Connector Only	C	Vision + Connector	D	Language + Vision
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{Results}



Baseline Model

Predicted Overall Score: 7

Explanation: The photograph **effectively uses monochrome** to emphasize texture and form. However, the image lacks dynamic contrast and compositional complexity to elevate it to a higher score.

Light: 6

Explanation: The lighting is **soft and diffused**, which helps to reveal the intricate textures of the wood without harsh shadows. However, the overall tonal range is somewhat muted, and the lack of strong directional light.

Color: 5

Explanation: Color is absent by design. But the grayscale palette **lacks vibrancy** or tonal richness.

Composition: 8

Explanation: The **composition is strong**, using a low-angle perspective and shallow depth of field to create a compelling focal point on the weathered wood.

Ground Mean
Score: 4.29

Calibrated Model

Predicted Overall Score: 4

Explanation: The image captures a sunset but **suffers from poor exposure**, rendering the foreground and midground with **little detail**. The sky's colors are visible but **muted and lack vibrancy**, and the composition is **unbalanced** with a distracting power line cutting across the frame.
Light: 5

Explanation: The scene is lit by the setting sun, but the overall **exposure is too low**, causing the foreground to be **underexposed** and details to be lost. The light is not effectively used to illuminate the scene or create contrast.

Color: 5

Explanation: The colors in the sky are present but appear washed out and **lack saturation**. The transition from orange to blue is visible but **not vivid**, and the dark foreground limits the color impact.

Composition: 4

Explanation: The composition is **weak** due to the power line cutting across the frame and the lack of a clear focal point. The road leads the eye but is **too dark** to be effective, and the horizon is not well-defined.

{Discussion & Conclusion}

What Worked



LoRA fine-tuning cut MAE by 57.6% over zero-shot baseline



Positive bias dropped from 1.602 to 0.150 (Config B)



Language-only LoRA (Config A) is strong, efficient, and stable



Structured prompt format achieved 0% parse failure across 400 images

What Didn't



Vision+Connector (Config C) collapsed – MAE jumped to 2.98



Score injection from the best performing CNN model into finetuned model showed negligible improvement over zero-shot



Zero-shot model compressed outputs to 6-8 range, failing on low-quality images



APDDv2 full dataset access still pending – limits cross-domain evaluation

{Discussion & Conclusion}

Next Steps



Scale dataset to full AVA (250k) and AADB (10k)



Cross-domain evaluation: train on photos, test on APDDv2 paintings

{References}

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{Thank You}

{Questions?}